

Tool runtime
emits a typed return and display text

$\tilde{z} = (s, x)$
validated typed return
residual binding uncertainty $B_{d,\varepsilon}(\tilde{z})$

m
display text · untrusted

LLM
reads $\tilde{z}$ and m · proposes action a

certified authorization boundary

CAGE monitor
reads $(\tilde{z}, a)$ · certifies all $z^* \in B_{d,\varepsilon}(\tilde{z})$

$\tilde{z}$ ✓ **certified**

a

m ✗ **not read**

Execute action a
authorized under $B_{d,\varepsilon}$

Abstain or escalate
unsafe or uncertain